Assessment Menu: Teacher Guide

The Slope Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
SLOPE	Find slope from points and data	A1, A2, E1, E6, S4
FORM	Slope-intercept form and graphing	A3, A4, E2, S3
WRITE	Write equations of lines	A5, E3, E4, S1
CONTEXT	Interpret slope and intercept in context	A6, E5, S2

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 SLOPE	Find the slope of the line through (2, 3) and (6, 11).	**Answer:** m = 2 m = $\frac{11-3}{6-2} = \frac{8}{4}$ m = 2	2 pts: Subtracts y-values and x-values (rise and run) 1 pts: Same order in numerator and denominator 2 pts: Correct slope 2
A2 SLOPE	The points (1, 9), (3, 5), and (6, −1) lie on one line. Find the slope of the line.	**Answer:** m = −2 Use (1, 9) and (3, 5): m = $\frac{5-9}{3-1} = \frac{-4}{2} = -2$ Check with (3, 5) and (6, −1): $\frac{-1-5}{6-3} = -2$ ✓	2 pts: Uses the slope formula with two of the points 1 pts: Same subtraction order top and bottom 2 pts: Correct slope −2
A3 FORM	Graph the line y = 2x − 1 on the grid.	**Answer:** A line through (0, −1) rising 2 units for every 1 unit to the right (also through (1, 1) and (2, 3)). The y-intercept is −1: plot (0, −1). The slope is 2 = $\frac{2}{1}$: from (0, −1) go up 2 and right 1 to (1, 1). Draw a straight line through the points.	2 pts: Plots the y-intercept (0, −1) 2 pts: Uses slope 2 to plot a second point 1 pts: Draws a straight line through the points
A4 FORM	Rewrite 2x + y = 8 in slope-intercept form and state the slope and y-intercept.	**Answer:** y = −2x + 8; slope −2, y-intercept 8 Subtract 2x from both sides: y = −2x + 8 The slope is −2 and the y-intercept is 8 (the point (0, 8)).	2 pts: Subtracts 2x from both sides 1 pts: Correct slope −2 2 pts: Correct y-intercept 8
A5 WRITE	Write the equation of the line with slope 2 that passes through (1, 5).	**Answer:** y = 2x + 3 Use y = mx + b with m = 2: y = 2x + b Substitute (1, 5): 5 = 2(1) + b, so b = 3 y = 2x + 3	2 pts: Substitutes the point into y = 2x + b 1 pts: Finds b = 3 2 pts: Correct equation y = 2x + 3
A6 CONTEXT	A candle's height in cm after *t* hours is h = 20 − 2t. In a sentence each, explain what the 20 and the −2 mean.	**Answer:** 20 is the starting height (cm); −2 means the candle burns down 2 cm every hour. The 20 is the height when t = 0, so the candle is 20 cm tall when lit (the intercept). The −2 is the rate of change: the height drops 2 cm for each hour (the slope).	2 pts: 20 = starting height (at t = 0), with units 3 pts: −2 = height decreases 2 cm per hour (decrease and rate stated)

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 SLOPE	The line through (1, k) and (5, 9) has slope $\frac{3}{2}$. Find k and show your work.	**Answer:** k = 3 Slope formula: $\frac{9-k}{5-1} = \frac{3}{2}$ So $\frac{9-k}{4} = \frac{3}{2}$; multiply both sides by 4: $9 - k = 6$ k = 3	3 pts: Sets up $\frac{9-k}{4} = \frac{3}{2}$ 3 pts: Solves $9 - k = 6$ 2 pts: Correct k = 3 2 pts: Checks the slope with k = 3
E2 FORM	Rewrite $3x - 4y = 20$ in slope-intercept form. State the slope, the y-intercept, and the x-intercept.	**Answer:** $y = \frac{3}{4}x - 5$; slope $\frac{3}{4}$, y-intercept -5, x-intercept $\frac{20}{3}$ Subtract 3x: $-4y = -3x + 20$ Divide every term by -4: $y = \frac{3}{4}x - 5$ x-intercept: let y = 0, so 3x = 20 and $x = \frac{20}{3}$	3 pts: Subtracts 3x and divides **every** term by -4 2 pts: Correct slope-intercept form 2 pts: Slope and y-intercept identified 3 pts: x-intercept found by setting y = 0
E3 WRITE	Write the equation of the line through $(-2, 7)$ and $(4, -5)$ in slope-intercept form.	**Answer:** $y = -2x + 3$ $m = \frac{-5-7}{4-(-2)} = \frac{-12}{6} = -2$ Use $(-2, 7)$: $7 = -2(-2) + b$, so b = 3: $y = -2x + 3$	4 pts: Correct slope -2 3 pts: Finds b = 3 using a point 3 pts: Correct equation $y = -2x + 3$
E4 WRITE	Write the equation of the line that passes through $(-1, 5)$ and is **parallel** to $4x - 2y = 6$.	**Answer:** $y = 2x + 7$ Solve for y: $-2y = -4x + 6$, so $y = 2x - 3$. The slope is 2. Parallel lines have equal slopes, so m = 2. Use $(-1, 5)$: $5 = 2(-1) + b$, so b = 7: $y = 2x + 7$	3 pts: Finds the slope 2 of the given line 2 pts: States parallel lines share the slope 2 pts: Finds b = 7 3 pts: Correct equation $y = 2x + 7$
E5 CONTEXT	A phone plan has a one-time activation fee plus a fixed monthly charge. After 2 months Ben has paid \$110 in total; after 5 months, \$215. Find the monthly charge and the activation fee, and write an equation for the total cost C after m months.	**Answer:** Monthly charge \$35, activation fee \$40; $C = 35m + 40$. Slope: $\frac{215-110}{5-2} = \frac{105}{3} = 35$, so the monthly charge is \$35. Substitute (2, 110): $110 = 35(2) + b$, so b = 40 (the activation fee): $C = 35m + 40$	3 pts: Slope 35 found from the two data points 2 pts: Interprets 35 as the monthly charge 3 pts: Finds the intercept 40 (activation fee) 2 pts: Equation $C = 35m + 40$
E6 SLOPE	Tara found the slope through $(-2, 5)$ and $(4, -7)$ like this: $m = \frac{-7-5}{-2-4} = \frac{-12}{-6} = 2$. Find her mistake, give the correct slope, and say what the sign of the correct slope tells you about the line.	**Answer:** She subtracted the x-values in the opposite order from the y-values. Correct slope -2: the line falls from left to right. Tara subtracted $y_2 - y_1$ on top but $x_1 - x_2$ on the bottom. The order must match. Correct: $m = \frac{-7-5}{4-(-2)} = \frac{-12}{6} = -2$ Negative slope: the line goes **down** as x increases (falls left to right).	4 pts: Identifies the inconsistent subtraction order 3 pts: Correct slope -2 3 pts: Explains a negative slope means the line falls left to right

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 WRITE	Line L passes through $(-4, 1)$ and $(2, 4)$. **(a)** Find the slope of L. **(b)** Write the equation of the line **perpendicular** to L through $(6, -3)$. **(c)** Where does your line cross the x-axis?	**Answer:** (a) $\frac{1}{2}$ (b) $y = -2x + 9$ (c) $(\frac{9}{2}, 0)$ (a) $m = \frac{4-1}{2-(-4)} = \frac{3}{6} = \frac{1}{2}$ (b) Perpendicular slopes are negative reciprocals: $m = -2$. Use $(6, -3)$: $-3 = -2(6) + b$, so $b = 9$: $y = -2x + 9$ (c) Set $y = 0$: $2x = 9$, so $x = \frac{9}{2}$ (the point $(\frac{9}{2}, 0)$)	4 pts: Slope of L is $\frac{1}{2}$ 4 pts: Perpendicular slope -2 (negative reciprocal) 4 pts: Finds $b = 9$ with the given point 4 pts: Correct equation $y = -2x + 9$ 4 pts: x-intercept: sets $y = 0$ and gets $\frac{9}{2}$
S2 CONTEXT	A tank drains at a steady rate. Its depth is 42 cm after 3 minutes and 27 cm after 8 minutes. **(a)** Find the slope. **(b)** Write an equation for the depth d after t minutes. **(c)** Explain what the slope and the intercept mean. **(d)** When will the tank be empty?	**Answer:** (a) -3 (b) $d = -3t + 51$ (c) Drains 3 cm per minute from 51 cm. (d) $t = 17$ minutes. (a) $m = \frac{27-42}{8-3} = \frac{-15}{5} = -3$ (b) $42 = -3(3) + b$, so $b = 51$: $d = -3t + 51$ (c) Slope -3: depth falls 3 cm per minute. Intercept 51: starting depth. (d) $d = 0$: $0 = -3t + 51$, so $t = 17$ minutes.	4 pts: Slope -3 4 pts: Equation $d = -3t + 51$ 4 pts: Interprets the slope with units (falls 3 cm per minute) 3 pts: Interprets the intercept (starting depth 51 cm) 5 pts: Sets $d = 0$ and finds $t = 17$ minutes
S3 FORM	Graph $2x + 3y = 12$ on the grid. **(a)** Find both intercepts. **(b)** Find the slope. **(c)** Draw the line. **(d)** Is (3, 2) on the line? Show how you know.	**Answer:** (a) (6, 0) and (0, 4) (b) $-\frac{2}{3}$ (d) Yes: $2(3) + 3(2) = 12$. (a) $y = 0$: $2x = 12$, so $x = 6$. $x = 0$: $3y = 12$, so $y = 4$. Intercepts (6, 0) and (0, 4). (b) Slope $= \frac{4-0}{0-6} = -\frac{2}{3}$ (or solve for y: $y = -\frac{2}{3}x + 4$) (c) Draw the line through (6, 0) and (0, 4). (d) $2(3) + 3(2) = 12$ ✓, so (3, 2) is on the line.	4 pts: x-intercept (6, 0) 3 pts: y-intercept (0, 4) 4 pts: Slope $-\frac{2}{3}$ 5 pts: Line correctly drawn through both intercepts 4 pts: (d) Substitutes (3, 2) and concludes it is on the line
S4 SLOPE	The points $A(-2, 1)$, $B(1, k)$, and $C(4, 7)$ lie on one line. **(a)** Find k using slopes. **(b)** Write the equation of the line. **(c)** Explain why the slope between A and B must equal the slope between B and C.	**Answer:** (a) $k = 4$ (b) $y = x + 3$ (c) A line has one constant slope. (a) Slope of AC $= \frac{7-1}{4-(-2)} = \frac{6}{6} = 1$ Slope of AB $= \frac{k-1}{1-(-2)} = \frac{k-1}{3}$ must equal 1, so $k - 1 = 3$ and $k = 4$. (b) Slope 1: $y = x + b$. Using (4, 7): $7 = 4 + b$, so $b = 3$: $y = x + 3$. (c) A line has the same steepness everywhere, so any two points give the same slope.	5 pts: Slope of AC = 1 5 pts: Sets slope AB = 1 and solves $k = 4$ 4 pts: Equation $y = x + 3$ 2 pts: Finds $b = 3$ by substituting a point 4 pts: (c) Explains a line has one constant slope